

August 2, 2026

RE: "Leveraging host omic-response signatures to identify microbiome metabolizers of drugs in the vaginal and gut microbiome" by D Nemecio *et al.*

To the Editorial Board and Editors of *Pacific Symposium on Biocomputing (PSB)*:

Enclosed, please find our research article entitled "Leveraging host omic-response signatures to identify microbiome metabolizers of drugs in the vaginal and gut microbiome," which we wish to submit for consideration for the *PSB* session "Systems Biology and Network Analysis: From Multi-omics Integration to Causal and Mechanistic Understanding." Our study generates a machine learning model that uses a microbe-drug-host (MDH) signature to predict specific microbe-drug metabolism in the vaginal and gut microbiome. The microbe-drug-host (MDH) signature integrates disease-free human omics, microbiome composition, and drug-treated *in vitro* gene expression data. This method presents a novel *in silico*, tissue-specific method to identify microbe-drug metabolic interactions which can impact clinical therapeutic success.

Microbiomes can negatively interact with exogenous drugs to impact their toxicity and reduce treatment efficacy. In the vaginal microbiome, tenofovir, a pre-exposure prophylaxis drug used to prevent HIV transmission is metabolized by microbes in the vaginal microbiome leading to a loss of host protection efficacy. The gut microbiome has known biotransformation with certain drugs, but species-level identification is rare. In this study we establish the pipeline to generate the SVMs using the MDH signature in a single-drug context with tenofovir in the vaginal microbiome. The vaginal models with MDH signatures based in transcriptomics and proteomics both reflect high performance and provide novel prediction on additional microbes. Using this framework we scaled up to a multidrug model in the gut microbiome using 130 drugs and 27 microbes. These models also contained reliable performance and the microbes most implicated to metabolize several of these drugs include *Bacteroides cellulosilyticus*, *Bacteroides thetaiotaomicron*, and *Blautia hansenii*. With this study we provide novel microbes predicted to metabolize drugs in both the vaginal and gut microbiome. This method addresses deficiencies of existing drug-microbe identification methods by providing a disease-agnostic, tissue-specific method to identify microbe-drug metabolism interactions.

We believe this work will be of strong interest to the diverse participants and readers of *PSB*. The manuscript aligns closely with the "**Systems Biology and Network Analysis**" session through its integration of host transcriptomics, microbiome composition, and pharmacologic perturbation signatures to perform microbe-drug metabolic prediction. **We confirm that this work is original, has not been published elsewhere, and is not under consideration by another journal or conference. All authors have approved the manuscript and agree with its submission to the *Pacific Symposium on Biocomputing*. We further disclose that generative AI tools were used to assist with coding during the computational analysis; all analyses, interpretations, and conclusions were reviewed and verified by the authors.**

Thank you for considering our manuscript.

Sincerely,



Douglas K. Brubaker, Ph.D.

Leveraging host omic-response signatures to identify microbiome metabolizers of drugs in the vaginal and gut microbiome

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The human microbiome interacts with drugs and contributes to side effects and loss of efficacy. Methods to identify strains with potential to biotransform drugs are inefficient, often fail to scale, have species-level specificity, and rarely account for host factors. We present an *in silico* method to predict the microbial metabolism of drugs using a Microbe-Drug-Host (MDH) signature. The signature integrates non-treated human omics, microbiome composition, and drug-treated *in vitro* transcriptomics to capture features that predict metabolic potential of a given microbe-drug pair. We built support vector machines (SVM) using MDH signatures to predict drug metabolism by vaginal and gut microbes. We established the framework with a single-drug vaginal test case and predicted metabolism of tenofovir with models with high performance metrics (AUC, precision, accuracy, recall, specificity equal to and above 0.99). Generalizing to multidrug gut models, 27 drugs across 130 bacteria were used to generate metabolism probabilities with overall high performance metrics with AUC, precision, accuracy, specificity above or equal to 0.81 and recall at 0.75. The gut microbes most implicated to metabolize drugs include *Bacteroides cellulosilyticus*, *Bacteroides thetaiotaomicron*, and *Blautia hansenii*, consistent with existing literature. Our method using the MDH signature addresses deficiencies of existing methods by providing a disease-agnostic, tissue-specific method to identify microbe-drug metabolism interactions.

Keywords: machine learning, microbiome, drug metabolism, host-microbiome interactions

1. Background

The microbiome dynamically contributes to host health outcomes in both beneficial and unsupportive manners. The gut microbiome aids in digestion of dietary fibers, synthesizes micronutrients, and assists with bile metabolism and creates a barrier of protection against pathogens.^{1–14} In the vaginal microbiome, bacterial strains provide protection against pathogens by maintaining a low pH environment through lactic acid production and releasing anti-microbial compounds such as hydrogen peroxide and bacteriocins.^{15–17} Microbiomes can biotransform exogenous drugs through reduction or oxidation reactions, resulting in side effects on the host such as toxicity or reduced treatment efficacy.^{18,19} These interactions have been best characterized in the gut microbiome.^{20–23} For instance, sulfasalazine requires an azo bond cleavage facilitated by gut microbiota-secreted azoreductases to become activated.^{24,25} Digoxin has a well-established inactivating metabolic relationship with *E. lenta* strains that possess a two-gene cytochrome-encoding operon.^{26–29} In the vaginal microbiome, the efficacy of the HIV pre-exposure prophylaxis drug tenofovir (TV) is compromised through metabolism by *Gardnerella vaginalis*, *Prevotella (bivia and amnii)*, *Mobiluncus mulieris* and *Escherichia coli* found to increase HIV incidence rate

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to 61%, compared to 18% in *Lactobacillus*-dominant microbiomes.^{30 31} In both the gut and vagina many drugs are metabolized by the whole microbiome, but few explicit bacterial strains have been identified. This remains a blind-spot for pharmacology towards comprehensively understanding the impact the microbiome has on drug therapies.^{19,32,33}

High-throughput methods for identifying microbe-drug interactions are extremely limited. Experimental methods use *ex vivo* and *in vitro* whole microbiome and single-species culturing with drugs. Computational methods for inference scale more effectively, but existing tools primarily predict *human* enzymatic interactions with drugs, and rarely predict *bacterial* metabolic capacity.^{34–38} Models that predict drug-microbiome interactions primarily explore whole-microbiome meta-metabolome output, or use functional metagenomics screens to identify bacteria carrying enzymes with known reactions.^{39–42}

In silico models specific to microbial metabolic products primarily use physiologically based pharmacokinetic (PBPK) models and constraint-based reconstruction and analysis (COBRA).⁴³ Vaginal models that predict bacterial metabolic capacity primarily use the metabolome to explore host-microbiome interactions using genome-scale metabolic models and flux balance analysis methods.^{34–37} These methods explore and predict metabolic outputs of the dynamic microbiome and may include omic data from host tissue to account for host-microbiome interactions. Other models explore prediction of drug metabolic product from bacteria in the context of therapeutics.^{44,45} Most models identify whole-microbiome biotransformation and few predict on species-specific drug metabolic capacity.

Here, we develop a computational framework using host, microbiome, and *in vitro* drug screening data to predict species-specific metabolism of drugs across multiple microbiomes using a novel “microbe-drug-host (MDH)” signature. Each microbe-drug pair contains a unique MDH signature that integrates the human, microbe, and reference drug data into a set of class-separating features that best reflects the metabolic relationship. We establish the framework with a case study of vaginal microbiome metabolism of TV and generalize the pipeline to a multidrug classifier for species-specific drug metabolism in the gut microbiome. This method takes advantage of the non-disease state of human *in vivo* omics combined with high-throughput *in vitro* drug transcriptomics to create a scalable framework to support large-scale screening and prediction for clinically meaningful microbe-drug metabolic relationships.

2. Materials and Methods

2.1. Pipeline

Fig. 1 depicts overall data preprocessing, feature selection, testing and training, and performance.

2.1.2. Data pre-processing

We generated a matrix that contains the integrated host omics, microbiome composition, and *in vitro* drug-treated transcriptomics information. Human transcriptomics or proteomics and paired microbiome composition undergo a Spearman correlation to produce a microbe-host ρ matrix. These ρ values are multiplied by each drug’s Library of Network-based Cellular Signatures characteristic direction (LINCS CD) to produce a product matrix. The CD is a mean gene expression over many cell types and treatment conditions. The point-wise product combines drug-treated *in vitro* data with *in vivo* microbe-host ρ . The relationship of a microbe-drug pair is denoted in the model with a binary class, +1 indicates a metabolic and -1 a non-metabolic relationship.

Microbe-drug pairs without literature-established relationships, denoted class of NA, are considered unknown and excluded from the training and testing data. Any microbe-drug pairs with

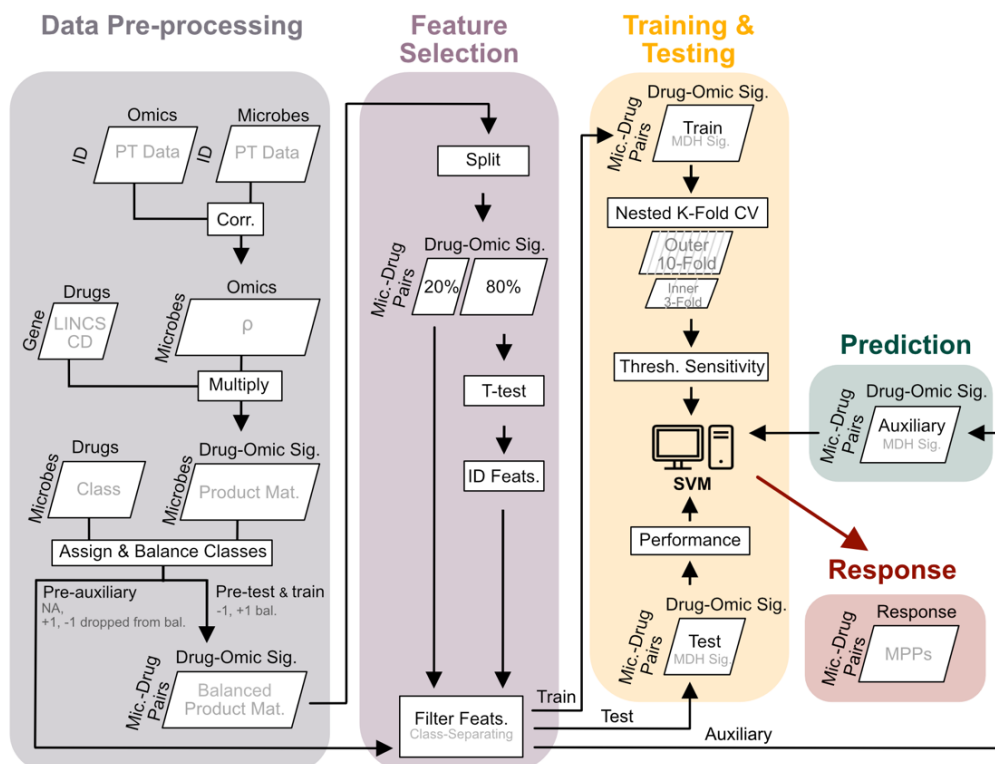


Fig.1: Generic pipeline to generate the MDH signature and SVM model to predict the metabolic relationship of the microbe-drug pairs. The Tv-VagGene, TV-VagProt and MultiDrugV1 and V2 models are adapted from this pipeline.

unknown metabolic class are sequestered to the novel predictions dataset which will be used for microbe-drug pair predictions after the model is generated.

2.1.3. Feature selection of the MDH signatures

We split the product matrix into testing and training data (split 20/80% respectively). The training data undergoes feature selection against the classes associated with the microbe-drug pairs via T-test with Benjamini-Hochberg false discovery rate (FDR) correction. Features with the lowest adjusted p-values define which features compose the MDH signatures for model training.

2.1.4. Training and testing

We used 10-fold cross validation (CV) with nested 3-fold hold out for hyperparameter optimization and performance metrics including precision, accuracy, recall, and specificity for each CV fold. Overall performance uses a prediction threshold of probability 0.5, where ≥ 0.5 is metabolic and < 0.5 probability is non-metabolic. We performed sensitivity analysis to determine the optimal threshold for each metabolic class at 0.05 increments. Optimal prediction thresholds were selected based on best F1 score for metabolic prediction responses and negative predictive value for non-metabolic prediction responses, with recall and specificity used to resolve ties respectively. If ties are not resolved, the lowest threshold for the metabolic class is selected and the highest threshold for the non-metabolic class is selected.

2.1.5. Predictions and response

Novel predictions are made with the optimized model using microbe-drug pairs that were excluded from the testing and training data. We assessed the prediction responses of the model independently on the testing and training data in addition to the auxiliary predictions. Predictions contain metabolic prediction probabilities (MPPs) for each microbe-drug pair based on the respective MDH signature. Prediction responses are finalized based on the optimal prediction thresholds determined from the sensitivity analysis. MPPs and the corresponding microbe-drug pairs that fall between the metabolic and non-metabolic classification thresholds are considered indeterminate.

2.2 Vaginal Data Participant Cohort

The vaginal models used HIV negative vaginal swab samples from the Partner's PrEP study.^{30,46,47} The TV-VagGene cohort contains 20 woman (25-47 years old) with 27 microbes represented and 9451 genes in vaginal tissue transcriptomics. (Supp. Fig. 1A) The TV-VagProt cohort contains 192 woman (18-49 years old) with 150 microbes represented and 434 identified proteins from host mucosa. (Supp. Fig. 1C)

2.3 Gut Data Participant Cohort

The gut models use 10 control, non-IBD adults from the Inflammatory Bowel Disease Multiomics Database (IBDMDB) as part of the Integrative Human Microbiome Project.⁴⁸ In total, 19 samples of ileum and rectum tissue transcriptomics with corresponding microbiome abundance was used. In the microbiome composition, 27 of the microbes were also found in the *in vitro* data, which will assign the microbe-drug "true" relationship for the testing and training data. (Supp. Fig. 1B) This is further described below in the 2.4.2 *Preparing Zimmerman Data* section.

2.4 Adapted Pipelines

The pipelines to generate the four models follow the same steps of data preprocessing, feature selection, testing and training, prediction and response. The vaginal models add bootstrapping and a modified feature selection. The gut model uses Zimmerman et al 2019 *in vitro* data from a microbe-drug screen to define which microbe-drug pairs to include in the testing and training data.⁴⁹

2.4.1 Vaginal Models Pipeline

Pre-processing with Bootstrapping

The vaginal models, TV-VagGene and TV-VagProt, undergo modified data preprocessing steps that include bootstrapping as depicted in Supp. Fig. 2A. The vaginal cohorts did not contain enough microbes with literature-established metabolic relationships of TV to create a balanced training and testing data set. We generated balanced testing and training data via bootstrapping. The participant omics data and microbe composition ρ were bootstrapped for 100 iterations with 10 participants selected for each iteration to produce a single microbe-host ρ matrix. The microbe-host correlations were multiplied by the LINCS CD to generate the product matrix. Metabolic relationships of TV were based on *in vitro* validated metabolic capabilities defined by Klatt et al 2017.³⁰ Microbes with metabolic capacity for TV include *G. vaginalis*, *M. mulieris*, *P. amnii*, *P. Bivia*, and *E. coli*. Microbes with non-metabolic capacity for TV include *L. crispatus* and *L. iners*. These classes are balanced and then filtered to exclude microbes with unknown metabolic capacity.

Feature Selection

Each of the 10 data frames underwent feature selection, where those with lowest 500 adjusted p-values were selected. The same features found across all 10 data frames were chosen as the MDH signature. This allows for identification of *consistent* features that distinguish the metabolic relationships *independent* of the noise introduced with bootstrapping. The first data frame is arbitrary chosen to be split 20/80% into testing and training partitions respectively.

2.4.2 Gut model's *in vitro*-defined relationships

The literature-established metabolic relationships needed for building the gut models is defined by *in vitro* data published by Zimmerman et al 2019.⁴⁹ This group assessed drug-metabolic activity of 76 human gut bacteria against 271 oral drugs, and reported percentage of drug consumed for each microbe-drug pair. The preprocessing is outlined in Supp. Fig. 2B. For our application, this *in vitro* data must be filtered to contain the same drugs and microbes found in the LINCS CD and IBDMDB cohorts respectively. The resulting filtered *in vitro* Zimmerman matrix contains 27 microbes shared with the IBDMDB cohort, and 130 drugs shared with LINCS CD.

Microbe Proportion and Percent Consumed Thresholds

We apply to the prepared *in vitro* Zimmerman matrix the *microbe proportion threshold* and *percent consumed threshold*. These thresholds impose cut-offs on the percent consumed values to in order to assign metabolic classes to the microbe-drug pairs. The first threshold is the *microbe proportion threshold*, which specifies for each drug the percent of microbes with a nonzero value of drug consumed. For example, a threshold of 0.25 means 25% of the 27 microbes tested with a drug contained *some* drug consumed. Any drugs and their respective microbes that do not meet this threshold are assigned a metabolic class of NA, and are dropped from the testing and training data.

The *percent consumed threshold* is applied to the remaining drugs and microbes, which specifies the percent of drug consumed to be considered metabolized by the respective microbe. For example, a threshold of 0.50 means 50% or more of the drug must be consumed to be considered a metabolic relationship and assigned a “true” class of +1, otherwise it is denoted as non-metabolic and assigned a class of -1. Each microbe-drug pair in the filtered *in vitro* Zimmerman matrix will have a metabolic class assigned based on the *percent consumed threshold*. The metabolic relationships are either metabolic (+1 class), non-metabolic (-1 class), or unknown (NA class). Microbe-drug pairs with 1) unknown metabolic relationships and 2) pairs dropped due to class balancing, compose the auxiliary data and are used for prediction at the end of the pipeline.

Multi-GutV1 model contains *microbe proportion threshold* of 0.50 and *percent consumed threshold* of 0.50. After sensitivity analysis was conducted on these thresholds, Multi-GutV2 contains *microbe proportion threshold* of 0.25 and *percent consumed threshold* of 0.50. For each gut model the testing, training, and auxiliary data together account for 3510 unique combinations of microbe-drug pairs derived from 27 taxa and 130 drugs.

3. Results

3.1 Vaginal Models' Performance

Feature selection identified 369 and 46 features for the TV-VagGene and TV-VagProt model respectively. The hyperparameter optimization determined the TV-VagGene model to contain the following parameters: kernel = polynomial, degree = 1, cost=10.2, and gamma=0.105. The hyperparameter optimization determined the TV-VagProt model to contain the following parameters: kernel = radial, degree = 1, cost = 10.3, and gamma = 0.451. The performance metrics

determined from the testing and aggregated CV training folds for both models are comparable and reflect useful classification from both methods. (Fig. 2C and Supp. Fig. 3A/B, 7A) Notably, precision, accuracy, recall and specificity are all above 0.99 from the testing fold and above 0.98 for the aggregated CV folds (Supp. Fig. 4) Both models reflect an area under the curve (AUC) of 1.00 in their respective testing and aggregated CV fold performances. The recall performance indicates that both models can identify metabolic microbes within the dataset (0.97, 0.99 for TV-VagGene and TV-VagProt respectively). Similarly, the high specificity indicates reliable identification of non-metabolic prediction responses (0.97 TV-VagGene, 0.99 TV-VagProt). The learning curve for both models also reflects negligible overfitting, as the generalization gap between the accuracy and mean misclassification error are less than 0.01 for both models. (Supp. Fig. 5)

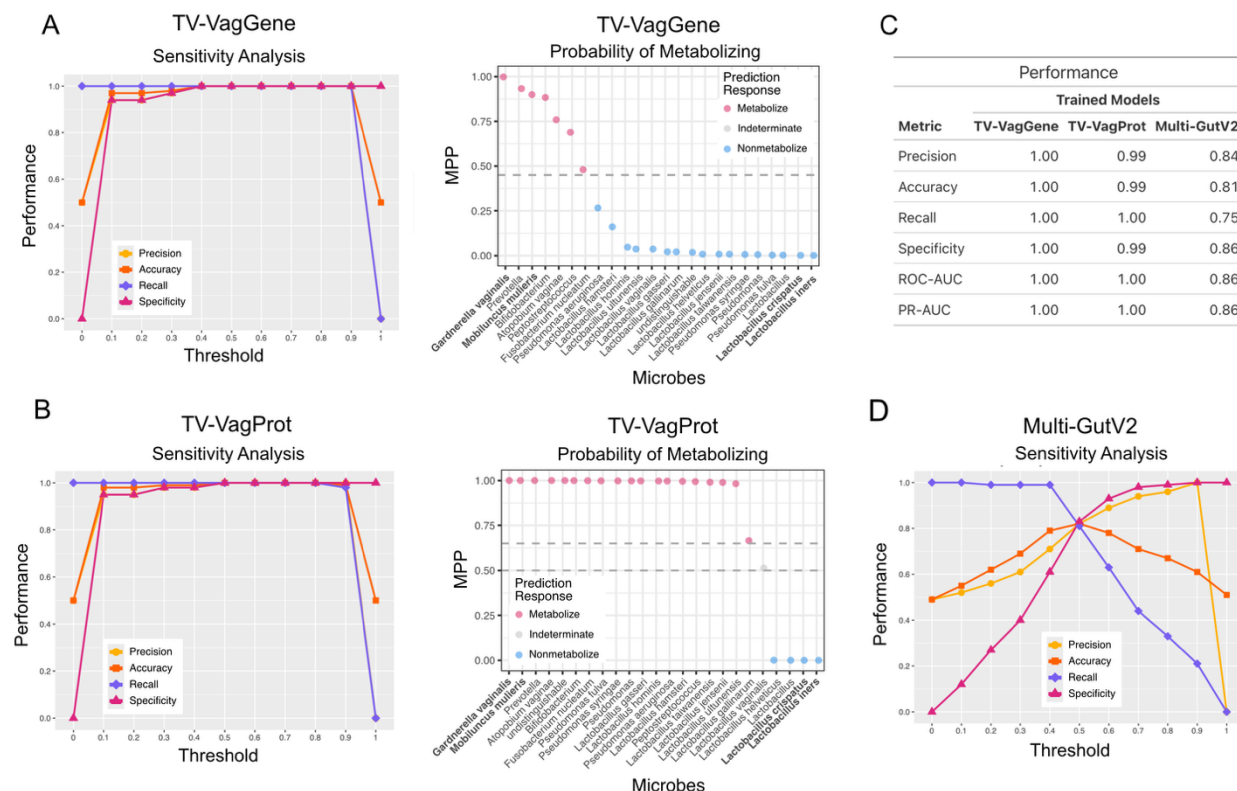


Fig. 2: Vaginal model results and performance metrics from TV-VagGene, TV-VagProt, and MultiDrug-V2. The sensitivity analysis on the classification threshold (A, B left) are determined from the test data. The MPPs (A, B right) for all direct participant MDH signatures are depicted for each model. Microbes found in both transcriptomics and proteomics data are displayed in the plots. Bolded microbes were used in testing and training. Complete predictions are found in Supplementary Tables 1 and 2. Optimized prediction thresholds determined from the sensitivity analysis are denoted as horizontal dashed lines. Performance metrics from three models are depicted (C). The sensitivity analysis on Multi-GutV2 classification threshold for aggregated CV fold reflects optimal performance at 0.45 (D) .

3.2 Vaginal Model Sensitivity Analysis

Sensitivity analysis on the classification threshold in both TV-VagGene and TV-VagProt reflect minimal impact on the overall performance. (Fig. 2A/B and Supp. 3) The TV-VagGene model optimal prediction threshold for response is 0.45, with an associated F1 value of 0.98 and recall of 0.98. The non-metabolic prediction threshold is also 0.45 with a determined negative predictive value of 0.99 and specificity of 0.97. The TV-VagProt model optimal metabolic prediction threshold is 0.65 with an associated F1 value of 1.00 and recall of 0.99. The nonmetabolic

prediction threshold is 0.5 with an associated negative predictive value of 1.00 and specificity of 0.99.

3.3 Vaginal Model Predictions

TV-VagGene and TV-VagProt model prediction responses on various microbes reflect a range of metabolic and non-metabolic responses. (Fig. 2 A/B) Both MPP plots reflect predictions in shared microbes from the TV-VagGene and TV-VagProt cohorts, all microbe MPPs are in Supp. Table 1 and 2 respectively. As expected, microbes used for training are predicted correctly with MPPs that support the response. Microbes predicted to metabolize TV includes *G. vaginalis* (1.00 MPP, 1.00 MPP), and *M. Mulieris* (1.00 MPP, 0.90 MPP) for TV-VagGene and TV-VagProt respectively. TV-VagProt also predicted metabolic response for *P. amnii*, *P. Bivia*, and *E. coli* microbes all with 1.00 MPP. Prediction response of nonmetabolic class is as expected for *L. Crispatus* (0.002 MPP, 1.00e-7 MPP) and *L. iners* (0.001 MPP, 1.00E-7 MPP) for TV-VagGene and TV-VagProt respectively. Comparatively, TV-VagGene returns more conservative predictions across all microbes, despite this, there are consistent predictions between both models.

Novel metabolic predictions shared by both models include *A. vaginae* (0.76 MPP, 1.00 MPP), Bifidobacterium species (0.88 MPP, 1.00 MPP) and Prevotella species (0.93 MPP, 1.00 MPP) from TV-VagGene and TV-VagProt respectively. Notably, Bifidobacterium and Atopobium genus are highly predicted to be metabolizers of TV. Due to the lack of availability in the cohort data, the TV-VagGene model was not trained on *P. amnii* and *P. bivia*; However, the model correctly identifies the metabolic potential of Prevotella genus. This supports the integrity of the method as the identification of this genus is correct, despite its absence from the training data.

The two models agreed on novel non-metabolic microbes which include *L. Helveticus* and Lactobacillus genus. There are nine microbes with conflicting predictions, primarily where TV-VagProt response predicts metabolic relationship (MPP range 0.98-1.00) and TV-VagGene does not (MPP range 0.003 to 0.16). These include *P. fulva*, *P. syringae*, Pseudomonas genus, *L. gasseri*, *L. hominis*, *L. hamsteri*, *L. taiwanensis*, *L. jensenii*, and *L. ultunensis*. A summary of the predictions between the TV-VagGene and TV-VagProt models is depicted in Supp. Fig. 3E.

3.4 Multi-GutV1

The results from Multi-GutV1 model with the *microbe proportion* threshold of 0.5 and *percent consumed* threshold of 0.5 depicts decent performance. (Supp. Fig. 6) The testing fold performance depicts a range of 0.75 - 0.77 for precision, accuracy, recall and specificity. (Supp. Fig. 6A) The AUC of the sensitivity and specificity curve and precision-recall curve are 0.83 and 0.87 respectively. These performance values and AUC support moderate performance by the model, and its ability to distinguish the metabolic class. (Supp. Fig. 6C) The sensitivity analysis on the prediction threshold reflects a 0.5 threshold is optimal for moderate performance across metrics. (Supp. Fig. 6B) The model's predictions grouped by drug reflect higher mean MPPs compared to grouping by microbe. (Supp. Fig. 6D/E) This reflects that MPPs are more influenced by the drug type rather than microbe species.

3.5 Multi-GutV2 Performance

During Multi-Gut model development we interrogated *microbe proportion threshold* and *percent consumed threshold* to identify a model with improved performance. Denoted as Multi-GutV2, this model contains a *microbe proportion threshold* of 0.25 and a *percent consumed threshold* of 0.50. This improved Multi-GutV2 model contains performance metrics of precision, accuracy,

recall, and specificity between 0.75-0.86. (Fig. 2 C, Supp. Fig. 7B) The learning curves reflect some minor potential overfitting with a generalization gap of 0.21. (Supp. Fig. 5) The model built on a balanced dataset is able to differentiate between the classes well (accuracy 0.81, specificity 0.86) and most positive predictions are correct (precision 0.84). This provides reassurance the prediction responses may provide meaningful MPPs worth interpreting. Sensitivity analysis on the prediction threshold reflects optimal performance at prediction threshold of 0.45 (F1 0.79, recall 0.83). (Fig. 2D) The optimal performance for nonmetabolic class is prediction threshold of 0.25 (negative predictive value 0.9, specificity 0.31). Performance metrics and confusion matrices for the aggregated CV folds are depicted in Supp. Fig. 3C/D.

3.6 Multi-GutV2 Trends in Responses

The Multi-GutV2 responses from the testing and training data show metabolic trends when MPPs are grouped by drugs rather than microbes. When grouped by microbes, the two leading microbes with the highest mean MPPs are *B. cellulosilyticus* and *B. thetaiotaomicron* (mean MPP 0.58, mean MPP 0.55). (Supp. Fig. 7C, Table 3) Interestingly, when the same MPPs are grouped by drug identity, certain drugs are favored for metabolic relationships by multiple species of microbes. (Fig. 3A, Table 3) The leading drugs with highest mean MPPs are artemisinin and tenatoprazole (mean MPP 0.88, mean MPP 0.84). 37% (18/49) of drugs contain mean MPPs that are above the metabolic prediction threshold of 0.45. There are no structural or pharmaceutical classes overrepresented in these 18 drugs. However, some of these drugs share metabolic mechanisms, specifically involving hydrolysis and reduction reactions. These 18 include prodrugs that require activation and contain the respective mean MPPs which include: bisacodyl (0.76), sulfasalazine (0.61), famciclovir (0.59), racecadotril (0.56), deflazacort (0.51). These also include known drug substrates for reduction and cleavage with the following mean MPPs: sulfasalazine (0.61), phenazopyridine (0.62), and tinidazole (0.75).

The predictions on the testing and training microbe-drug pairs reflects consistent classification of the model's response and the corresponding true class. For microbe-drug pairs assigned a class "truth" of metabolic, the median MPP is 0.68 and median percent consumed is 0.99. For the "true" non-metabolic microbe-drug pairs, the median MPP is 0.12 and median percent consumed is 0.0. A Spearman correlation between the MPPs for all testing and training microbe-drug pairs and the corresponding percent consumed *in vitro* value returns a ρ of 0.55 with a p-value of $1.3e-53$. The positive direction, meaningful magnitude and notable p-value reflect the *in vitro* percent consumed values associated with each microbe-drug pair are highly correlated to the MPP produced by the model. This correlation reflects the MDH signature can capture wide metabolic range, and corresponding MPP's are reflective of measured activity in the *in vitro* values used to assign the classes. Together, these values support the useful classification abilities of the model.

Compared to the testing and training MPPs, the predictions on the auxiliary microbe-drug pairings assigned a truth of NA reflect a smaller range of MPPs. This is evident when grouping the MPPs by microbe and by drug. The two leading microbes identified to metabolize the drugs include *B. hansenii* (mean MPP 0.50) and *B. thetaiotaomicron* (mean MPP 0.50). (Supp. Fig. 7D , Supp. Table 3) The two leading drugs predicted to be metabolized by the microbes are metaxalone (mean MPP 0.64) and fenofibrate (mean MPP 0.54). (Fig. 3B) 44% (36/81) of the drugs contain mean MPPs above the metabolic prediction threshold of 0.45. The auxiliary microbe-drug pairs were selected based on failure to meet the *microbe proportion threshold* and *percent consumed threshold*, and therefore it is reasonable that we see more MPPs trending towards non-metabolic

responses. Similar to the testing and training predictions, the auxiliary microbe-drug pair responses are more distinguishable based on the drug rather than the microbes. Experimentally these auxiliary microbe-drug pairs had very little metabolism representation across the microbes. The MPPs are less certain compared to those produced by the testing and training data, but are still above our classification threshold for metabolic predictions.

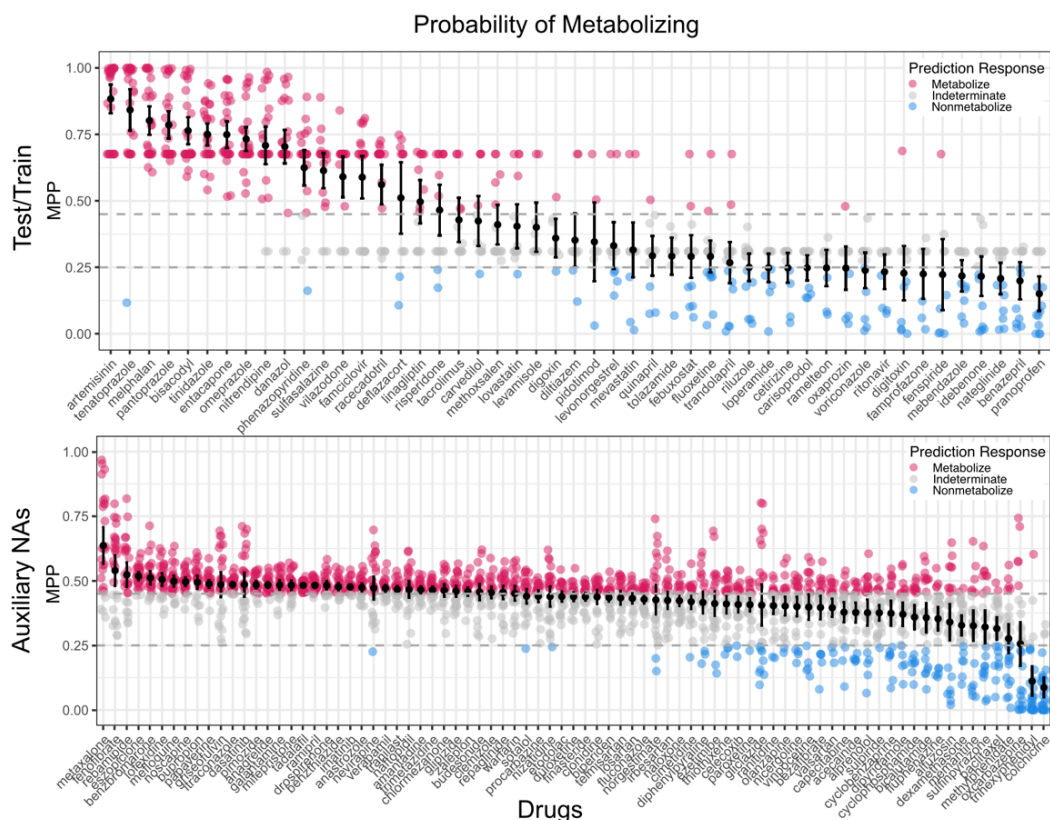


Fig. 3: Multi-GutV2 prediction responses grouped by drugs from the training and testing data (top) and auxiliary NA data (bottom). Optimized prediction thresholds determined from the sensitivity analysis are denoted as horizontal dashed lines.

3.7 Literature Exploration of Multi-GutV2 Responses

We assess Multi-GutV2 responses with literature-established gut microbiome metabolic relationships to explore our results. These include microbe-drug pairs from across the testing, training and auxiliary data. (Table 3) We explore known relationships between sulfasalazine, omeprazole, digoxin and levamisole metabolism evidence. The sulfasalazine literature-established relationship with the gut microbiome is reflected in the MPPs. Microbe-sulfasalazine pairs predicted as metabolic returned a relatively high median MPP of 0.68. For sulfasalazine, 78% (21/27) of microbes are predicted as metabolic and 22% (6/27) are indeterminate. Interestingly, for these six the true assigned class is denoted as non-metabolic, but since the MPPs are not low enough meet the non-metabolic prediction threshold of 0.25, they fall within the indeterminate class. This suggests the associated MDH signatures for these six sulfasalazine-microbe pairs still contain enough “metabolic” signature so that the model is unable to confidently predict them to be non-metabolic. Our results reflect *P. distasonis* and *B. vulgatus* contain high MPPs (0.89, 0.68 respectively). Interestingly, *ex vivo* human fecal cultures treated with sulfasalazine displayed a significantly increased species abundance of these same two bacterial species.⁵⁰ Additionally, of

the *Bacteroides* genera, *B. caccae* and *B. Fragillis* (MPP 0.77, MPP 0.75) are most likely metabolic towards sulfasalazine. Within literature, *B. caccae* has been identified as a biomarker for children with ulcerative colitis.⁵¹ At the same time, *B. caccae* is also reported as a predictor for successful treatment outcomes with anti-cytokine therapies.²⁰ There are no findings associating successful treatment outcomes with sulfasalazine in children with ulcerative colitis and *B. caccae* abundance. However, the *Bacteroides* genera has been identified to contain azoreductases implicated in activating sulfasalazine through reduction.⁵² Indeed our predictions reflect this trend, where 83% (10/12) of the *Bacteroides* species are predicted to be metabolic of sulfasalazine with a median MPP of 0.68. There are two *Bacteroides* species that are denoted as undetermined, *B. stercoris* and *B. finegoldii*, however the *in vitro* assigned truth is non-metabolic. The lack of consistency between the truth and response may be due to metabolic capacity being captured in the MDH signature. Reductive metabolism of omeprazole has been reported in *in vitro* conditions by gut bacteria to produce sulfide metabolites however there are few species explicitly identified.^{26,53} In our model, 96% (26/27) of microbe-omeprazole pairs were assigned a truth of metabolic and 100% (27/27) had a predicted response of metabolic. Digoxin has an interesting literature established relationship with *E. lenta*, where only strains that contain a two-gene cytochrome-encoding operon metabolize the drug. Our results (MPP 0.31, indeterminate) reflect this range of *E. Lenta* response, since it is not a metabolic relationship held by the entire species and thusly the MPP is low enough to be placed in the indeterminate response. Due to the MDH signature capturing some of this metabolic capacity the MPPs is not below the non-metabolic threshold. Microbe-levamisole pairs had responses that deviated from the “truth” assigned. There are two microbes *A. colihominis* and *E. coli* out of the 27 levamisole-microbe pairs assigned a truth of metabolic. Interestingly, the model prediction responses identified seven microbes to be metabolic (median MPP 0.53). Evidence suggests Levamisole may be metabolized by *Bacteroides* species, and indeed three of these levamisole-microbe pairs included *B. uniformis* (MPP 0.72), *B. cellulosilyticus* (MPP 0.54), and *B. thetaiotaomicron* (MPP 0.50).⁵⁴ Our results with sulfasalazine, omeprazole, digoxin and levamisole are explored due to their existing literature evidence to have metabolic properties with the gut microbiome. These results highlight the model’s capability to return predictions that align with literature-based findings. Discrepancies between the response and truth may indicate the MDH signature’s ability to discern between the *in vivo* metabolic capacity independent of the assigned “true” class based on the *in vitro* data.

Within the auxiliary prediction data, we explore the prediction responses for microbe-drug pairs with an unknown true class. (Fig. 3B, Supp. Fig. 7B, Table 3) Some of the 18 drugs identified to be metabolized include capecitabine, sulfinpyrazone, ranitidine and nizatidine. Our results reflect varying degrees of model responses that align with literature-established evidence. Capecitabine is reduced by dihydropyrimidine dehydrogenase, an enzyme previously identified in 85 species of bacteria through comparative genomics.⁴¹ In our results, capecitabine returned 22% (6/27) microbes as having a metabolic relationship (median MPP 0.49). Sulfinpyrazone can experience s-oxide reduction by microbes found in human stool.^{55,56} For sulfinpyrazone, 30% (8/27) microbes were predicted to have metabolic relationship (median MPP 0.52). Ranitidine and nizatidine, when incubated with human stool experience a nitro reduction via cleavage of an N-oxide bond.^{57,58} Ranitidine-microbe pairs had 33% (9/27) predicted to be metabolic (median MPP 0.47). Nizatidine-microbe pairs had 52% (14/27) microbes predicted to metabolic (median MPP 0.51). Despite these microbe-drug pairs being sequestered to the auxiliary data during preprocessing class assignment due to their inability to meet the *percent consumed* and *microbe proportion* threshold, we still see predictions for microbes with metabolic capacity. The median

MPPs are lower compared to those from the testing/training. This is somewhat reasonable, since the auxiliary data was not used in training, and therefore the MDH signature is not tuned to detect any metabolic capacity within these drugs. Despite this, the metabolic predictions demonstrate the generalizable, MDH signature *can* distinguish metabolic capacities on drugs that did not contribute to the training of the model.

Metabolic relationships identified from *in vitro* screening with drugs can contribute to our understanding of drug-microbe interactions. However, it does not capture dynamic microbiome composition which often impacts bacterial metabolic need and activity. While we can state what the assigned “truth” is based off the *in vitro* data, the endogenous behavior of the bacteria species in the host microbiome context may be different. Due to this misalignment, it is possible that behavior observed *in vitro* is not behavior observed in the endogenous host microbiome. These results reflect the MDH signature derived from *in vivo* patient data can capture metabolic relationships, which are difficult to replicate *in vitro*. The MDH signature improves on the existing *in vitro* and *in silico* methods as it integrates the *context* of the complete endogenous microbiome interaction with the host. The Multi-GutV2 model has identified species-specific microbes culpable for metabolic activity with various drugs, which were previously unknown.

4. Discussion

The MDH signature can predict species-specific metabolic relationships. It is derived from human omics data and microbiome composition from disease-free subjects and integrates the drug-treatment variable through *in vitro* derived data. Our findings highlight the novelty and utility of using a MDH signature to identify drug-microbial metabolic relationships in the vaginal and gut microbiome. Using this MDH signature we built vaginal microbiome models, TV-VagGene and TV-VagProt, that predict metabolism of TV. These models contained high performance metrics on test data (precision, accuracy, recall, specificity equal to and above 0.99) and an AUC equal to and above 0.99. The two models agreed on the novel predictions of TV metabolism by the vaginal bacteria: *A. vaginae*, *F. nucleatum*, *Peptostreptococcus* genus, and *Bifidobacterium* genus.

We further demonstrated the greater capacity of the MDH signature to capture multiple microbe-drug metabolic relationships in the gut microbiome with the Multi-GutV1 and Multi-GutV2 models. The Multi-GutV2 model was identified through sensitivity training of the V1 model and contained useable performance metrics from the testing data (precision, accuracy, specificity above or equal to 0.81 and recall at 0.75), with an AUC of 0.87. For the Multi-GutV2 model the leading microbes most implicated in metabolism of the assessed drugs are *B. cellulosilyticus*, *B. thetaiotaomicron*, and *B. hansenii*. There are 54 total drugs with mean MPPs across the respective microbe-drug pairs assed that reflect metabolic potential with multiple microbe species. The responses also reflect trends in drug literature-established metabolic relationships. The Multi-GutV2 model produced MPPs in the testing and training data which are highly correlated (ρ of 0.55, p-value $1.3e-53$) to the percent consumed values from the *in vitro* data. This reflects the MDH signature’s ability to capture and discern the range of metabolic capacity of each unique microbe-drug pair.

The method to produce the MDH signatures is highly customizable as it requires 1) non-disease paired omics and 2) a few microbe-drugs with defined metabolic relationships. It is advantageous the human omics and microbiome composition is *non-disease* as this expands the range of applicable available datasets. The human paired omics and microbiome composition are tissue-specific and disease-agnostic. The training data requires defined metabolic relationships for microbe-drug pairs to establish the model. These can be taken from literature (as shown in TV-

VagGene and TV-VagProt models) or custom *in vitro* experiments with a threshold to delineate defined metabolic relationships (as done in the Multi-GutV1 and V2 models).

Future applications of this model could adapt the existing pipeline and shift the specificity of the model to suit the research interests. The utility of this model is constrained to the microbiome diversity of patient samples. This can impact the performance of the model though limited samples for testing and training or limit the amount of novel microbe-drug pairs that are predicted on. Future applications of this method could generate models unique to specific drug classes or hypothesized enzymatic mechanisms. Drugs in similar classes or with similar functional groups have been shown to share metabolic properties consistent across different microbes with the same enzymes.^{19,32,33,49,59} Similar pathways may be involved in shared structures and therefore the MDH signature may identify related features. Structure could be delineated by the molecule composition, such as core ring systems (pyrimidines or thiazoles), functional groups (amides, guanidines) or a super structure (polycyclic aromatics, macrocycles).

Another improvement could be applying tissue-specific *in vitro* drug-treated cell line data rather than the LINCS CD values. The LINCS CD readings include a vast diversity of cell types across various tissue locations. The models may have better performance if the *in vitro* values were more tissue-specific and therefore would have more contextually accurate drug-treatment values to contribute to the MDH signatures. Future applications of this model could also expand using a larger applied drug pool. Here, we limited our microbe-drug pairs to drugs that were found in both our *in vitro* drug-treated microbes from the Zimmerman et al 2019 publication to simplify interpretation of our results and to ensure we had *in vitro* values to compare our generated MPP per microbe-drug pair. Future applications could use the whole LINCS catalogue to generate microbe-drug pair predictions for all LINCS drugs to generate a reference database. This generated library of predictions could be cited to preemptively identify any potential impact of the gut microbiome on the drug of interest. We could also make individual SVMs per drug for each microbiome, similar to the individual models we made for TV metabolism. These single-drug models may contain higher precision since the class-separating features in the MDH signatures capture each specific drug's metabolic properties. The weakness of a multidrug model such as the Multi-GutV1 and V2 is the class-distinguishing features are *not* specific to the drug class nor the structures, which can contribute to loss of performance. This is reasonable, since introducing additional drugs may obscure the features that best distinguish the classification abilities. However, despite this, in a multidrug model the features are more generalizable *because* the features are not specifically tied to one drug. We have shown reliable performance is still viable in a multidrug model.

The utility of these models is most evident within the clinical setting. We can pre-emptively determine how a patient's microbiome may impact the therapeutic effect of the drug. With this knowledge we can either pivot drug treatment to select a therapeutic that will not interact with the microbiome or provide probiotics to shift the microbiome populations to be more receptive to the select drug of choice. This is a low-cost, high throughput method to assess the microbe metabolism of drugs in an accessible manner. Clinicians can reference this tool to incorporate microbiome state into patient treatment plans going forward, thereby improving personalized medicine and treatment success.

5. Supplementary Information and References

All supplementary figures, tables and references can be found here: https://github.com/Brubaker-Lab/microbe-host-omic_nemecio

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